Progress Report

- Increment 2 -

Group Metroidvania

1) **Team Members**

Skyler Haas, smh23a, 149013192

Raul Medrano, rem22r, 196469754

Armani Ruiz, arj22c, 158619431

Evan Treadwell, eat24c, 247679165

John Wibert, jnw22c, 183773469

2) **Project Title and Description**

Our project is currently titled “A Bone to Pick”, which will be a short Metroidvania-style video game in which players can control a character from a 2D side scrolling perspective to explore an interconnected world, with a focus on smooth movement and tight, responsive controls. By exploring the map and defeating enemies scattered around the world, players can collect various items and upgrades to help them traverse the world, unlock new pathways, and aid in combat with enemies and the final boss. The game will be designed to be beatable in one sitting.

3) **Accomplishments and overall project status during this increment**

Level design has been the main focus during this increment. This means that an updated and fully functional tileset was implemented, including different types of ground, ceiling, and wall textures, along with decorative objects to make the world feel alive. The levels were designed with the sole purpose of picking up the necessary items needed to progress throughout the game, while also providing a fair challenge for the player. Along with this, new enemy types such as a flying enemy and a “pouncing” enemy were implemented, along with their respective designs. We have also changed the way health operates in the game, aiming towards a heart point-based system rather than a percentage system. The overall status of the project has been on pace with what our scope entails.

4) **Challenges, changes in the plan and scope of the project and things that went wrong during this increment**

There was a slight error when importing the pouncer enemy onto the correct scenes, where the animations wouldn’t load due to an extra instance of the object being created where it relies on a separate parent object. To fix this, we simply added the code for the animations onto the newly created instance of the pouncer enemy where it doesn’t rely on a parent object. There have been slight changes to how the levels are going to be designed, more-so aiming at a non-linear approach that is more in-line with the genre we’re going for rather than simply just platforming. We have also thought of creating bonus sections for the player to navigate, aimed to further upgrade the player character with extra powerups, such as a low-gravity powerup. The scope of the project has maintained the same.

5) **Team Member Contribution for this increment**

Skyler Haas: Responsible for constructing and narrating the video demonstrating the Progress Report with guidance/oversight from other members. Responsible for making sure the "Non-functional requirements" section of the RD doc is kept updated with the non-functional requirements of our project such as control responsiveness, level readability, and reliability. Provided oversight and input for the IT doc, primarily the "Execution-based Non-Functional Testing" section as it related heavily to his work on the RD doc.

Contributions to the source code for this increment primarily include building the various areas and levels of the game.

Raul Medrano: Responsible for ensuring the first four parts of the Progress Report (listing the team members, providing a basic description of the project, describing accomplishments and project status for this increment, and describing the challenges and scope changes during this increment) are kept updated. Responsible for making sure the "Functional requirements" section of the RD doc is kept updated by listing the functional requirements of our project such as the interconnected map and player moveset requirements. Provided oversight and input for the IT doc, primarily the "Execution-based Functional Testing" section as it related heavily to his work on the RD doc. Contributions to the source code include designing and implementing hand-drawn sprites and animations for the player character, enemies, and other level assets, as well as experimenting with other graphical elements such as tilesets and parallax scrolling for the background. Provided some input and oversight for the video, namely providing sprites and gameplay clips for Skyler to show off as demonstrations of the team's progress.

Armani Ruiz: Responsible for writing the "Team Member Contribution for this increment" section of the Progress Report, listing each team member and describing their contributions to each deliverable. Responsible for ensuring the "Operating Environment" and "Assumptions and Dependencies" sections of the RD doc are kept updated, laying out the game engine and system requirements for our project as well as the assumptions and dependencies that inform and guide the development of the project overall. Responsible for ensuring the IT doc is kept updated using the inputs provided by other team members, with primary focus on updating the "Non-execution-based Testing" section. Highly focused on maintaining the team's Scrum-inspired workflow by keeping record of each team member's tasks and issues for each sprint. Contributions to the source code include designing and refining some enemy behavior as well as assisting Skyler's work by providing some research, ideas, and expertise for level design. Provided some input and guidance on the video including helping to decide what to include on each slide.

Evan Treadwell: Responsible for writing the "Stakeholder communication" section of the Progress Report, drafting an email describing our progress that would be sent to the publishing team/company for our game if we hypothetically wanted to publish it. Also responsible for ensuring the "Class diagram and/or sequence diagram" section of the RD doc is kept updated, designing an overview of how different aspects of our system's architecture, such as UI/HUD elements, enemies, and the player interact with each other within the game. Provided oversight and input for the IT doc, primarily the "Programming languages" section. Contributions to source code include ongoing development of various UI and HUD elements for the game, such as the pause menu and player health, as well as valuable work on a save/load feature. Provided some input and guidance on the video including helping to decide what to include on each slide.

John Wibert: Responsible for writing the "Plans for the next increment" section of the Progress Report, describing some ideas for what elements the team will focus on going forward such as level design, UI elements, and the final boss. Responsible for ensuring the "Use Case Diagram" section of the RD doc is kept updated by designing a simple use case diagram to show how a player may interact with the game's elements and systems on a surface level. Provided oversight and input for the IT doc, namely the "Platforms, APIs, Databases, and other technologies used" section. Contributions to the source code include finalizing the player character's movement feel and controls, and implementing/refining the unlockable abilities that will be key to progression in the finished game. Worked closely with Skyler to develop an outline for the video.

6) Plans for the next increment

In the next increment, we plan to take the tools we've developed and fine-tuned to design the levels the player will traverse throughout the game. We spent a lot of time creating and tuning enemies, tilesets, obstacles, and player abilities in order to make this next increment as smooth sailing as possible. We also want to finish the design of the UI and menu systems, alongside designing the final boss the player will fight at the end of the game. With the majority of the project's most backend code completed, we should hopefully be able to build the next components without needing to design new systems from the ground-up.

7) **Stakeholder Communication**

Hi Publisher team,

Metroidvania has grown from a single test area into a set of drafted levels. Since our last update, we've built a tutorial and five early level layouts, and ability-based progression is now in place. The air dash and air attack are now earned through pickups in the world rather than available from the start, and a temporary low-gravity power-up adds variety to platforming.

The enemy roster has doubled from two types to four. Joining the heavy hitter and the fast attacker are a pouncer, which winds up and leaps at the player's position, and a flying enemy that patrols the air. Levels now also include environmental hazards such as spikes, and bounce mushrooms that reward skillful use of the mid-air attack.

On the systems side, the game now has a main menu with New Game and Continue options, and a save system that keeps the player's health, unlocked abilities and progress between sessions. The pause menu has also been expanded with saving, volume settings and a return to the main menu.

Map art is now in use across the new levels, and work continues on enemy and environment art.

Our next milestone focuses on connecting these levels into one continuous map, adding save stations as in-world checkpoints, and continuing to refine level design and enemy placement.

We're happy to share an updated playable build at your convenience.

8) **Link to video**
https://1drv.ms/p/c/cd75b58f6b4ed0f7/IQDJrOWFUSz2QZVr0bogRTHSAVxZSRSJFLOPCvXeH4rinDs?e=rK5dxU